

Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science Engineering

Sep 2022 – Jun 26

CGPA: 8.57/10

Manas Sthali Public School

Class XII – 83.8% in PCM

2021-22

Class X – 82.4%

2019-20

Internships

Betalectic IT Projects Pvt Ltd

Software Intern

May 2024 – July 2024

- Designed and deployed UIs leveraging ReactJS and Agile methodologies to deliver user-centric solutions with rapid iteration.
- Built and optimized analytics features with Laravel and RESTful APIs; enhanced application scalability through performance refactoring and CI/CD best practices.
- Improved the performance of key visualization components, helping reduce load times by approximately 20%.
- Actively participated debugging, gaining real-world experience in blending data analytics with web development.

Bharat Intern

Data Science Intern

Dec 2023 – Jan 2024

- Delivered a CNN image classifier and an SMS spam detector (98% accuracy) following best practices in MLOps.
- Used TensorFlow/Keras with NLP techniques such as tokenization, stop-word removal, TF-IDF for data preprocessing.
- Optimized the model through feature engineering, hyperparameter tuning, and validation on labeled datasets cross-validation, model deployment, and continuous evaluation.

Academic Projects

Cognitive Compliance Platform | [LIVE](#) • [Github](#)

- Built an AI-powered compliance platform to automate regulatory scanning, risk extraction, and LLM cross-verification.
- Leveraged advanced NLP (transformer-based NER, context extraction, risk detection) for structured legal insights, with Gemini LLMs ensuring cross-check accuracy.
- Architected an interactive dashboard with dynamic visualizations and reporting tools, allowing users to filter risks, track entities, and export compliance summaries as PDF reports.
- Achieved a robust solution with improved accuracy, strengthening compliance workflows and proactive risk mitigation.

AI Co-director | [LIVE](#) • [Github](#)

- Architected an Agentic AI-driven script transformation system using CrewAI, designing a 3-agent pipeline (Writer → Critic → Rectifier) that converts scripts into intended genres through iterative validation and refinement, for a real-world workflow.
- Engineered a multimodal emotion intelligence engine by fusing video, audio, and text models to verify whether a video's delivered emotion aligns with the intended narrative tone.
- Implemented an intelligent feedback loop where Generative AI provides actionable cinematic improvement suggestions (dialogue tone, pacing, sound design, visual intensity) when detected and intended emotions diverge.

Abnormal Activity Detection | [Github](#)

- Developed a deep learning pipeline for detecting abnormal human activities (e.g., burglary, abuse, arrest) from video input using a custom spatio-temporal model.
- Integrated CNN for feature extraction and RNN for temporal sequence modeling, enabling real-time multi-class classification.
- Pre-processed and trained on UCF Crime dataset, achieving 77% multi-class classification accuracy.
- Optimized performance through sequence-level augmentation, learning rate scheduling, and model regularization for deployment in edge surveillance systems.

Certifications

- Microsoft Certified: Azure AI Engineer Associate
- Google Cybersecurity Certificate
- AWS Academy Cloud Architecting
- Columbia+ certified: Prompt Engineering and Programming with Open AI

Skills & Interests

Technical Skills:

- Programming & Tools: Python, MySQL, Java, JavaScript, Git, Linux, Spark, Tableau, UI/UX, LLM, OpenAI SDK, LangChain
- Data Engineering: Data Validation, Data Cleaning, Debugging Workflows, Testing for Data Pipelines, Model Debugging.
- ML & Data Science: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, NLTK, SpaCy, Matplotlib, Seaborn, Pytorch
- Web Dev: Reactjs, Django, Laravel, HTML, CSS, REST APIs, FastAPI

Soft Skills: Problem Solving, Business mindset, Team Collaboration, Leadership, Time Management, Critical thinking, Adoption.